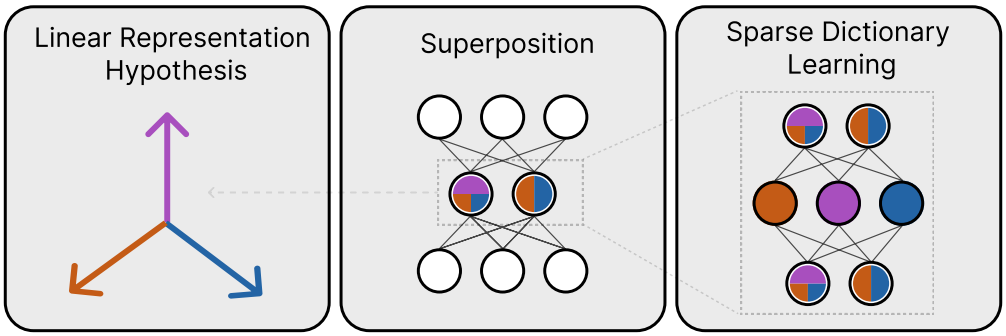

Figure 3: Three ideas underlying the sparse dictionary learning (SDL) paradigm in mechanistic interpretability. (Left) The linear representation hypothesis states that the map from 'concepts' to neural activations is linear. (Middle) Superposition is the hypothesis that models represent many more concepts than they have dimensions by representing them both sparsely and linearly in activation spaces. (Right) SDL attempts to recover an overcomplete basis of concepts represented in superposition in activation space.

input or, in the case of transcoders (Dunefsky et al., 2024), to reconstruct the activations of the next layer. Crosscoders (Lindsey et al., 2024) permit a wider class of inputs and outputs, potentially reconstructing the activations of many layers simultaneously. Since the encoder is nonlinear, it is thought to be able to learn to activate a latent only if a feature is 'active' in the hidden activations and remain 'off' if it is not.

Although SDL is considered a leading decomposition method for mechanistic interpretability, it has substantial practical and conceptual limitations (Figure **4**).

SDL reconstruction errors are too high: Large errors in SDL reconstruction raise the question whether SDL methods can reconstruct the hidden representation sufficiently well such that the latents learned by SDL are adequately faithful to the models being interpreted. To measure this, the model's true hidden activations can be replaced with sparse dictionary reconstructions, then subsequently evaluating the extent to which the model's performance decreases. In practice, the error results in significant performance reductions. When a sparse dictionary with 16 million latents was inserted into GPT-4, the language modeling loss was equivalent to a model with only 10% of GPT-4's pretraining compute (Gao et al., 2024). Similarly, Makelov et al. (2024) found that using reconstructions from sparse autoencoders decreased GPT-2 small performance by 10% when trained on task-specific data, and 40% when trained on the full distribution.

Making sparse dictionaries much larger and sparser to reduce errors is a feasible but computationally expensive approach. Furthermore, in the limit, this results in merely assigning one dictionary latent per datapoint, which is clearly less interpretable. One partial solution is using SDL methods with 'error nodes' (Marks et al., 2024), to account for the discrepancies between the original and reconstructed activations. However, while the sparse autoencoders identified interpretable latents, the error nodes contain 'everything else', making them an inadequate solution to the problem. Engels et al. (2024b) found that these reconstruction errors are not purely random, as much of the direction of the error and its norm can be linearly predicted from the initial activation vector. This suggests that current SDL methods may systematically fail to capture certain structured aspects of model representations, but also implies potential solutions. To overcome this issue, it may be necessary to improve SDL training methods, or develop entirely new methods of network decomposition.

SDL methods are expensive to apply to large models: SDL involves training a small neural network for every layer of the AI model that we want to interpret. Typically, the sparse dictionaries have more parameters at that layer than the original model does, and consequently will probably be relatively expensive to train compared to the original model. [3] As AI models become larger, scaling costs of SDL also increase, although it remains unclear whether relative scaling costs are sub- or supra-linear. For cost effectiveness,

[3]The actual relative cost is unclear since there are no public attempts to apply SDL to every vector space in a model, although some work applies SDL to various layers (Marks et al., 2024; Gao et al., 2024; Braun et al., 2024; Lieberum et al., 2024; Bloom & Lin, 2024a; Huben et al., 2024)

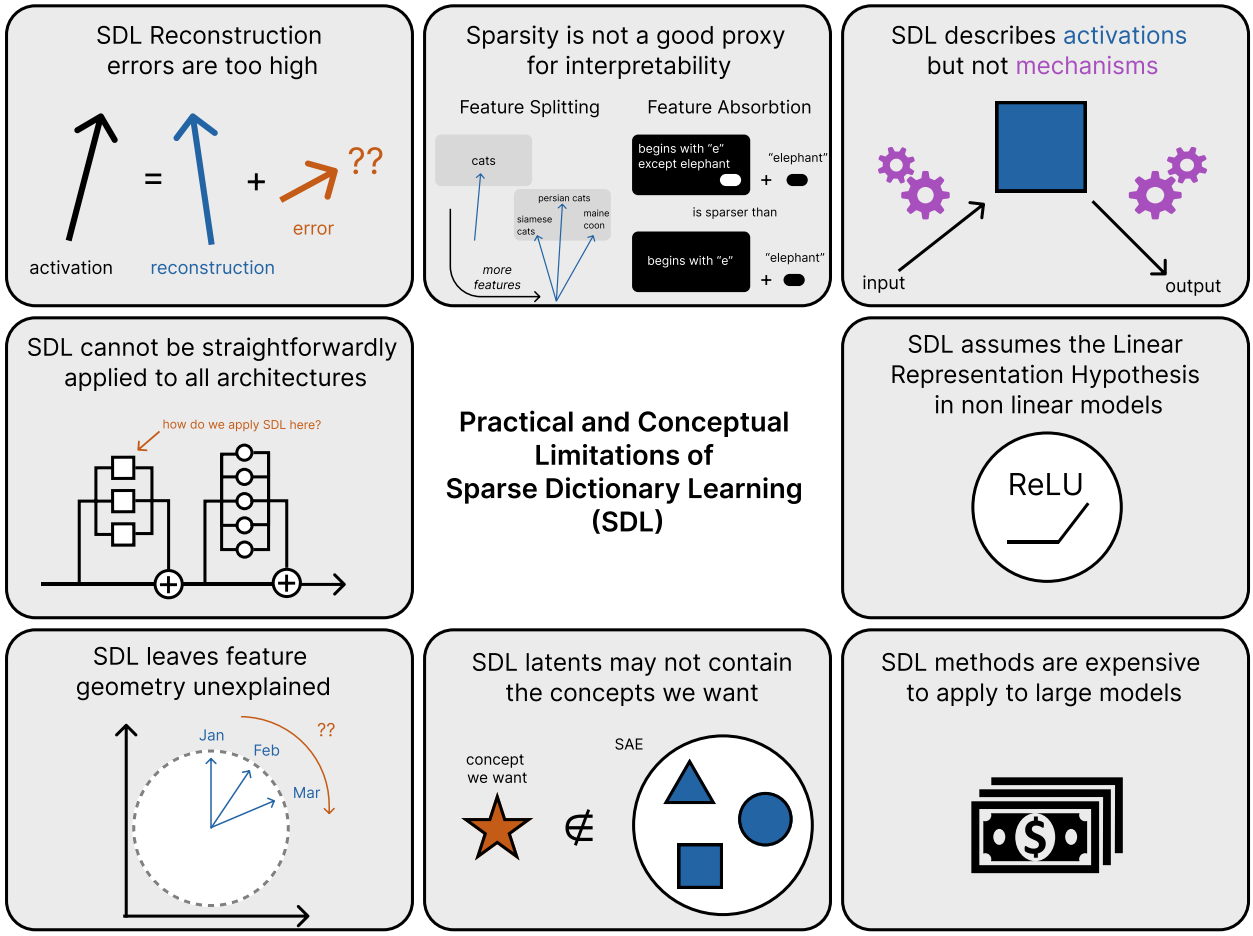

Figure 4: Sparse dictionary learning has a number of practical and conceptual limitations that cause issues when using it to reverse engineer neural networks (Paragraph 2.1.2c).

it may be important to develop intrinsically decomposable methods for training models, while remaining at (or close to) state-of-the-art performance (Paragraph 2.1.2e). This approach will help avoid incurring the expense of both training and decomposing AI models.

SDL assumes the linear representation hypothesis in nonlinear models: Other problems with SDL arise from the assumptions on which SDL is based. One such assumption is the linear representation hypothesis. The linear representation hypothesis observes that though neural networks are nonlinear functions and could potentially use highly nonlinear representations (Black et al., 2022; Engels et al., 2024b; Kirch et al., 2024), they tend to use representations that exhibit strikingly linear behavior (Smolensky, 1986; Mikolov et al., 2013; Olah et al., 2020b; Elhage et al., 2021; Park et al., 2023a; Guerner et al., 2024; Gurnee & Tegmark, 2024). The hypothesis formalizes this phenomenon by stating that high level concepts are linearly represented as directions (vectors) in neural network embeddings. Its two core claims are (i) that the composition of multiple concepts can be represented as the addition of their corresponding feature vectors, and (ii) that the intensity of a concept is represented by the scale of its corresponding feature vector (Olah & Jermyn, 2024). Earlier work (Elhage et al., 2021) defined the linear representation hypothesis with the additional assumption of one-dimensional features, but this definition has since been refuted (Yedidia, 2023; Chughtai & Lau, 2024; Engels et al., 2024a) and clarified (Olah & Jermyn, 2024). Another recently proposed criterion is that the intensity of concepts should be retrievable by a linear function of the embeddings, up to a small error (Hänni et al., 2024; Olah & Jermyn, 2024).

A weak version of the linear representation hypothesis states that some concepts are linearly represented, while a strong version may assert that all concepts are (Smith, 2024a). Some works have shown that the strong version is false for some models (Black et al., 2022; Csordás et al., 2024). The weaker version of the linear representation hypothesis is supported by the successes of linear probes (Section 2.2.1), activation steering (Section 3.2.2), and the success of sparse autoencoders in finding seemingly interpretable latents (Paragraph 2.1.2c).

Sparsity is not a good proxy for interpretability: Another of SDL's key assumptions is that feature activations are sparse. Therefore, SDL methods optimize their latents accordingly to be sparsely activating, with the implicit assumption that sparser decompositions are more interpretable than denser ones. However, this assumption may not necessarily hold. The (related) problems of feature splitting (Bricken, 2023), feature absorption (Chanin et al., 2024) and composition (Till, 2024) suggest that with sufficient optimization pressure, sparsity as a proxy for interpretability breaks down (though note that it is debated whether feature splitting is actually a problem). Other proxies, such as minimum description length, might be better optimization targets than sparsity *per se* (Ayonrinde et al., 2024). It also may be the case that no proxy metric is sufficient.

SDL leaves feature geometry unexplained: SDL decomposes networks into single directions in the activation space, which is a reasonable approach only if we consider feature activations akin to 'a bag of features' (Harris, 1954) without any internal structure. However, the geometric arrangement of features in relation to each other seems to reflect semantic and functional structure (Engels et al., 2024a; Gurnee & Tegmark, 2024; Park et al., 2024b; Bussman et al., 2024). Understanding the geometric structure of a network means understanding the position of one feature relative to (potentially) many others. Comprehending this may be necessary if we want to know why and how a network treats certain features similarly or differently. If understanding the global geometry (all-to-all relationships) of features is essential to understand neural networks, this might pose a fundamental problem for current approaches to mechanistic interpretability (Hänni et al., 2024; Mendel, 2024). However, if only local geometric relationships between features need to be understood, understanding networks with a 'bag of features' approach may be more feasible.

SDL cannot straightforwardly be applied to all architectures: SDL was originally developed to identify features that may be represented in superposition across many neurons within a single layer. However, representations may be spread over other network architecture components besides neurons. In transformers, for example, representations may be spread across separate attention heads (Jermyn et al., 2023; Janiak et al., 2023), and even across different layers (Yun et al., 2021; Lindsey et al., 2024). However, it is not immediately obvious how to decompose representations distributed across attention heads with SDL (Mathwin et al., 2024; Wynroe & Sharkey, 2024). Nor is it straightforward to translate cross-layer distributed representations into causal descriptions of neural network mechanisms (Lindsey et al., 2024).

SDL decomposes the input and output activations of network mechanisms, but not the mechanisms themselves: Ultimately, the aim of mechanistic interpretability is to understand the mechanisms learned by neural networks. The parameters of the network, along with its nonlinearities and other architectural components, implement these mechanisms, which are applied to the input's hidden activations. SDL identifies directions in activation space. Being activations, they only interact with the network's mechanisms, but are not the mechanisms themselves. Describing the mechanisms directly remains unresolved with SDL. Gaining insights about the network's mechanisms from SDL latents requires further post hoc analysis, which can be labor intensive, computationally expensive, or data set dependent (Huben et al., 2024; Bricken, 2023; Riggs et al., 2023; Marks et al., 2024). This is an instance of a more broad problem with current mechanistic interpretability work; we primarily focus on understanding neural network activations, with little attention paid to how this structure in activations is computed via weights (Chughtai & Bushnaq, 2025).

SDL latents may not contain the concepts needed for downstream use cases: When using SAEs for practical tasks, there sometimes exists a single latent or small set of latents representing some concept of interest for task performance (and not representing much else). For instance, Kantamneni et al. (2024) found a single latent whose activation pattern was more accurate than official dataset labels on the NLP task GLUE CoLA (Warstadt et al., 2019). More often than not though, a sparse set of latents that encode some useful concept of interest do not exist. It is unclear what causes this problem. One hypothesis is that the concept